



Indirect Treatment Comparisons of Mosunetuzumab With Third- and Later-Line Treatments for Relapsed/Refractory Follicular Lymphoma

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Abstract

ITCs were conducted to compare mosunetuzumab with other ≥ 3 L treatments for relapsed/refractory follicular lymphoma. The results suggested that mosunetuzumab is superior to tazemetostat (in patients without *EZH2* mutations) and PI3K inhibitors. For anti-CD20 mAb therapies, outcomes varied depending on the efficacy endpoint. Results versus CART therapies varied depending on the technology, although mosunetuzumab's accessibility and safety should also be considered.

Background: No established standard of care exists for relapsed/refractory (RR) follicular lymphoma (FL) after ≥ 2 prior therapies. We conducted indirect treatment comparisons (ITCs) to compare the efficacy and tolerability of mosunetuzumab with those of available treatments used in this setting. **Methods:** A systematic literature review (SLR) and subsequent feasibility assessments were conducted to identify the most suitable comparator studies in terms of design, available endpoints and populations. Imbalances in patient characteristics between NCT02500407 and studies featuring aggregate or patient-level data availability were accounted for using matching-adjusted indirect comparison (MAIC) and propensity score-based methodologies, respectively. **Results:** ZUMA-5, ELARA, DELTA, DYNAMO, UNITY-NHL, AUGMENT and NCT01897571 passed the MAIC feasibility assessment. Patient-level data were available from GADOLIN, CONTRALTO and NCT02257567. MAIC results generally favored mosunetuzumab over tazemetostat in *EZH2*wild-type patients for all outcomes and over PI3K inhibitors for complete response (CR), objective response rate (ORR), discontinuations due to adverse events and progression-free survival (PFS) with umbralisib. MAICs favored CART therapies for PFS and, to a lesser extent, ORR and CR. Comparisons with anti-CD20 antibody-based regimens yielded mixed results. **Conclusions:** ITCs suggest that mosunetuzumab may lead to superior outcomes over tazemetostat (in *EZH2*wild-type patients) and PI3K inhibitors and may be a promising alternative to re-challenging with a different anti-CD20 regimen in patients who relapse after ≥ 2 prior anti-CD20 lines. Although preliminary results somewhat favored CART therapies, limitations and uncertainties remain because of intrinsic differences in study design. Mosunetuzumab could thus be a promising treatment option for patients with RR FL after ≥ 2 prior therapies.

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Keywords: Anti-CD20 antibodies, Bispecific antibody, CART therapy, Comparative efficacy, Non-Hodgkin lymphoma

Abbreviations: 2L, second line; 3L, third line; AE, adverse event; ASCT, autologous hematopoietic stem-cell transplant; ATE, average treatment effect; BCa, bias corrected and accelerated; CART, chimeric antigen receptor T cell; CHOP, rituximab or obinutuzumab in combination with cyclophosphamide, doxorubicin, vincristine and prednisone; CI, confidence interval; CR, complete response; CVP, cyclophosphamide, vincristine and prednisone; DLBCL, diffuse large B-cell lymphoma; ECOG PS, Eastern Cooperative Oncology Group performance status; ESS, effective sample size; *EZH2*, enhancer of zeste homolog 2; FL, follicular lymphoma; FLIPI, Follicular Lymphoma International Prognostic Index; Hgb, hemoglobin; HR, hazard ratio; HTA, health technology assessment; HSCT, hematopoietic stem cell transplant; INV, investigator; IPD, individual patient-level data; IPTW, inverse probability of treatment weighting; IRF, independent review facility; ITC, indirect treatment comparison; LDH, lactate dehydrogenase; LLN, lower limit of normal; mAb, monoclonal antibody; MAIC, matching-adjusted indirect comparison; MCL, mantle cell lymphoma; MZL, marginal zone lymphoma; NE, not estimable; NHL, non-Hodgkin lymphoma; NICE, National Institute for Health and Care Excellence; OR, odds ratio; ORR, objective response rate; OS, overall survival; PET, positron emission tomography; PFS, progression-free survival; PI3K, phosphoinositide 3-kinase; PL, prior lines of therapy; POD24, progression of disease within 24 months; PSA, propensity score analysis; R-CHOP, rituximab,

cyclophosphamide, doxorubicin, vincristine, prednisone; RR, relapsed refractory; SD, stable disease; SLL, small lymphocytic lymphoma; SLR, systematic literature review; WT, wild type.

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Introduction

Follicular lymphoma (FL) is one of the most common types of non-Hodgkin lymphoma (NHL) and the second most common lymphoma diagnosed in Western Europe and the US.^{1,2} Despite substantial therapeutic progress with the use of immunochemotherapy (rituximab or obinutuzumab in combination with cyclophosphamide, doxorubicin, vincristine and prednisone [CHOP]; cyclophosphamide, vincristine and prednisone [CVP]; or bendamustine)³ as first-line treatment, most patients eventually relapse.⁴

No standard of care has been established for patients with relapsed refractory (RR) FL. As a result, treatment patterns vary widely.⁵ Second-line (2L) treatments include alternative, potentially non-cross-resistant immunochemotherapy, rituximab plus lenalidomide or anti-CD20 monotherapy. Phosphoinositide 3-kinase (PI3K) inhibitors were initially approved as third-line (3L) treatment for RR FL (idelalisib,⁶ duvelisib,⁷ umbralisib⁸ and copanlisib⁹), but most have subsequently been withdrawn in many countries by their manufacturers for this indication.¹⁰ Some of the emerging treatments recently approved by the US Food and Drug Administration and/or the European Medicines Agency to treat RR FL in 3L or later settings include the CD19-directed chimeric antigen receptor T-cell (CART) therapies axicabtagene ciloleucel¹¹ and tisagenlecleucel,¹² the enhancer of zeste homolog 2 (EZH2) inhibitor tazemetostat¹³ and the first T-cell-engaging bispecific antibody, mosunetuzumab.¹⁴⁻¹⁷

Mosunetuzumab is a CD20×CD3 T-cell-engaging bispecific monoclonal antibody that targets CD20 on B cells and CD3 on T-cells to redirect endogenous T cells to engage and eliminate malignant B cells.¹⁸ Preclinical studies showed that mosunetuzumab induced rapid and sustained T-cell activation and proliferation and had the potential to circumvent rituximab resistance with potent lysis of CD20-expressing B cells, including primary leukemia and lymphoma cells.¹⁸ In a pivotal Phase II study of mosunetuzumab monotherapy in patients with RR FL after ≥2 prior systemic therapies, the objective response rate (ORR) was 80% and the complete response (CR) rate was 60%¹⁴; the median duration of response was 22.8 months. The safety profile was manageable, with cytokine release syndrome being predominantly low grade and occurring mostly in Cycle 1.¹⁴

In the absence of head-to-head randomized controlled trials, indirect treatment comparisons (ITCs) provide an alternative method for comparing the relative treatment effects of different therapies by adjusting for between-trial differences in baseline patient characteristics.¹⁹⁻²¹

To determine how the efficacy and tolerability of mosunetuzumab would indirectly compare with those of other relevant therapies for FL in the 3L or later setting, we first conducted a systematic literature review (SLR) to identify and review the available evidence for existing standard-of-care and emerging treatments for RR FL. Thereafter, a feasibility assessment was conducted to identify the most suitable published studies for conducting ITCs to compare mosunetuzumab with specific key comparators of interest. Finally, a series of ITCs of efficacy and tolerability outcomes were made using matching-adjusted indirect comparisons (MAICs) (when individual

patient-level data [IPD] for a comparator study were not available) and propensity score analyses (PSAs) of mosunetuzumab versus treatment regimens of interest.

Methods

SLR

To identify all available evidence for standard-of-care treatments or novel emerging therapies for the management of RR FL in the 3L or later setting, a SLR was conducted as follows in accordance with PRISMA reporting guidelines (Supplemental Methods). The eligible patient population included adult patients with RR FL (or RR FL within a population of patients with other indolent NHL histologies) treated in the 2L setting or in the 3L or later.

Studies conducted in patients with RR FL who received ≥2 prior therapies were of primary interest, but the criteria were defined to include the broader ≥2L indolent NHL patient population to make sure studies that included multiple lymphoma histologies (hereafter referred to as “mixed lymphoma”) and/or treatment lines were not missed. A hierarchical approach was subsequently applied to ensure that only studies enrolling comparable populations were used in the analyses.

The study categorization criteria were as follows: ≥3L FL only, ≥3L mixed lymphoma, ≥2L FL only, ≥2L mixed lymphoma and 2L mixed lymphoma/FL. Studies of patients in the following settings were included: 1) ≥2L (with at least 50% ≥3L) FL, 2) ≥3L FL or 3) ≥3L with mixed lymphoma that reported baseline data and results for the FL subgroup. Studies of mixed lymphoma and line of treatment with no relevant subgroup data available were also included, but cutoffs were applied for the proportion of FL patients (>80% or ≤80% FL) and line (median ≥2 prior lines or median 1 prior line). Exceptions were considered to allow for inclusion of studies of widely prescribed treatments in 3L+ FL that borderline met these criteria.

With the exception of 2L mixed lymphoma/FL-only studies, all relevant studies of the key interventions of interest identified in the SLR were considered for data extraction and for the initial stage of the MAIC feasibility assessment (Supplemental Methods).

MAIC Feasibility Assessment

Following the SLR, the feasibility of each MAIC was assessed by evaluating cross-trial similarities and differences between the pivotal mosunetuzumab study GO29781¹⁴ and each comparator study regarding key inclusion/exclusion criteria, availability of information on prognostic factors and effect modifiers of relevance for the condition (Table 1) and the outcomes of interest reported: overall survival (OS), progression-free survival (PFS), ORR, CR and treatment discontinuation due to adverse events (AEs). Where multiple studies for a given comparator were deemed feasible for conducting a MAIC, only the most appropriate comparator study was chosen. As 5 of the available studies were sponsored by Roche, IPD were available to be used in PSAs as recommended by the National Institute for Health and Care Excellence (NICE)²¹ and were therefore not included in the MAIC feasibility assessment.

Table 1 Priority of Baseline Prognostic Factors Considered in ITCs

Priority	Prognostic/Predictive Factor	Definition (Measured at Study Entry)
High	No. of previous chemotherapeutic (or systemic) regimens	3 vs. >3 (no clinically established threshold) or median, if categories not reported
	Refractoriness to last previous therapy	Yes (defined as best response of disease progression or relapse <6 mo from last therapy) vs. no ^a
	Refractoriness to any prior anti-CD20 mAb-containing therapy	Yes vs. no ^a ; also used as proxy for rituximab refractoriness when needed
	Early relapse status (POD24)	Yes vs. no; could be defined differently across studies; definition: "from first treatment" was preferred over "from diagnosis," as aligned with the definition used in the GO29781 trial
	Prior (A)SCT	Yes vs. no
	Size of largest lymph node lesion involved	Highly correlated with bulky disease, hence was prioritized over bulky disease when possible
	Bulky disease	Yes vs. no ^b
	FLIPI risk group	High (≥ 3) vs. intermediate/low (<3) at study entry; if at diagnosis, this had limited value in $\geq 3L$ setting and individual components were to be prioritized
	Age	Mean, or median if mean not reported, or % aged ≥ 60 years (when feasible) if neither reported
	Ann Arbor Stage	I-II vs. III-IV
	High LDH	Yes vs. no
	Bone marrow involvement	Yes vs. no (as demonstrated by bone marrow biopsy) when available
	Low hemoglobin level	Yes vs. no; eg, <12 (or 12.5) g/dL or <LLN
Low ^c	Duration of prior response/time in previous remission/time since completion of last therapy	
	ECOG PS	0 vs. 1 (the distinction between 0 and 1 and ≥ 2 carries the greatest prognostic value)
	Presence of B symptoms	Not as important because these may be subjective

Abbreviations: 3L = third line; (A)SCT = (autologous) stem cell transplant; ECOG PS = Eastern Cooperative Oncology Group performance status; FLIPI = Follicular Lymphoma International Prognostic Index; ITC = indirect treatment comparison; LDH = lactate dehydrogenase; LLN = lower limit of normal; mAb = monoclonal antibody; POD24 = progression of disease within 24 months.

^a Double refractoriness to both a rituximab-containing regimen and an alkylating agent was flagged as of unclear priority but potentially of relevant prognostic value in identifying a potentially higher risk population; therefore, it was controlled for (when available) and tested in sensitivity analyses.

^b Bulky disease is generally constructed from the size of largest lymph node involved; as none of the thresholds typically used to define bulky disease have been established as being superior prognostically over the others (based on medical feedback), then adjusting for bulky disease in the MAICs should be de-prioritized in favor of size of largest lymph lesion when information on both is available.

^c Although serum beta 2 microglobulin level (high vs. low) was flagged as a potentially relevant prognostic factor, it was eventually not considered in the analyses, owing to the fact that it can often be confounded by other biomarkers not related to the disease, based on medical opinion. Also, it was not available for the $\geq 3L$ FL patient cohort in the mosunetuzumab trial.

Source Data on Mosunetuzumab

IPD for mosunetuzumab were available from the single-arm Phase I/II trial, GO29781 (NCT02500407),¹⁴ which included 90 patients with FL who had been exposed to the mosunetuzumab registrational dosing regimen after ≥ 2 prior therapies. The main inclusion criteria for the FL expansion cohort are listed in Supplemental Table 5.

All analyses were conducted using the data cutoff of August 27, 2021.

General Statistical Methodology

MAIC requires IPD from at least one of the trials being compared to re-weight an intervention's IPD to reflect patients' over- or under-representation with respect to prognostic factors and effect modifiers relative to the comparator study population. This allows estimation of the outcomes that would be observed in patients receiving the treatment if the population were the same as that enrolled in the comparator trial.^{20,21} When IPD are available for both studies, PSA methods are more suitable for providing an estimate of relative treatment benefits.

Known prognostic factors and effect modifiers for RR FL to be considered in the ITCs were classified as either high or low priority (Table 1) based on discussions with clinical experts. Handling of differing prognostic factor definitions and missing data

is described in the Supplemental Methods, as are the statistical software packages.

First, the populations in GO29781 and each comparator study were equalized by aligning the eligibility criteria related to the prognostic factors in Table 1, where appropriate and feasible. Outcomes of interest reported for GO29781 were matched to the available definitions for the comparator trials (Table 2), where feasible. Independent review facility (IRF)-assessed response and PFS endpoints were preferred over investigator-assessed endpoints when both were available. If IRF-assessed endpoints were not available, investigator-assessed endpoints from both studies were compared to ensure consistency. Time-to-event outcomes (ie, PFS and OS) were analyzed using the Kaplan-Meier method; treatment effect estimates were expressed as hazard ratios (HRs) and 95% confidence intervals (CIs) using a Cox proportional-hazards model (exact method was used for tie data). Binary outcomes (ie, ORR, CR and discontinuations due to AEs) were analyzed using logistic regression and odds ratios (ORs) with 95% CIs. CIs were estimated using bootstrapping with 2000 resamples.²²

Statistical software is described in the statistical methods.

MAIC Methodology

Unanchored MAICs were used because data for mosunetuzumab and most comparators were only available from single-arm studies.

Table 2 Overview of Comparator Studies Selected for MAIC Analysis

Comparator	Study, N	Key Inclusion Criteria	Population		Baseline Data Available	Outcome Reporting	Analysis Population	Key Efficacy Outcomes				
			Mixed FL n (%) FL	Line of Therapy N (%) ≥3L Median (Range PL)				OS Median (95% CI)	PFS Median (95% CI)	ORR n (%; 95% CI)	CR n (%; 95% CI)	Dis-continuation Due to AEs
CART therapy												
Axicabtagene ciloleucel	ZUMA-5 ^{11,23,24–26,a} N = 149	RR FL or MZL after ≥2 prior therapies ECOG PS 0-1 Patients with SD (without relapse) >1 year from last therapy completion were ineligible	Mixed 124 (83%)	≥3L 3 (2-9)	FL ≥3L	FL ≥3L	Efficacy (long-term analysis n = 86, FL patients with ≥9 mo follow-up)	Long-term analysis KM curve NR (39.6-NE)	Long-term analysis KM curve 39.6 (25.7, NE)	Long-term analysis 81 (94%) Lugano 2014 ²⁷	Long-term analysis 68 (79%) Lugano 2014 ²⁷	n/a
Tisagenlecleucel	ELARA ^{12,28–32,a} N = 97	RR FL grade 1 to 3a ≥2 prior therapies	FL Safety population = 97 (100%)	≥3L	FL ≥3L	FL ≥3L	Efficacy (n = 94, at least 6 mo postinfusion follow-up)	Extended follow-up analysis KM curve Median not reached	Extended follow-up analysisKM curve 29.9 (17.9, NE)	81 (86%; 95% CI, 78-92) ²⁷ Lugano 2014	65 (66%; 95% CI, 56-75) ²⁷ Lugano 2014	n/a
PI3K inhibitors												
Idelalisib	DELTA ^{6,33–36,b} N = 125	RR B-cell indolent NHL after ≥2 prior therapies Refractory to both rituximab and an alkylating agent	Mixed 72 (58)	≥3L	FL ≥3L	FL ≥3L	All patients who received ≥1 dose of study drug	KM curve 61.2 (38.1, NE)	KM curve 11 (8, 14)	40 (55.6%; 95% CI, 43.4-67.3) Cheson 2007 ³⁷	12 (16.7%) Cheson 2007 ³⁷	17 (23.6%)
Duvelisib	DYNAMO ^{7,38,b} N = 129	RR FL, SLL or MZL Refractory to rituximab and chemotherapy or radio-immunotherapy ECOG PS 0-2	Mixed 83 (64)	≥2L 112 (87)	FL ≥2L	FL ≥2L ≥3L (ORR)	Full analysis set (all patients who received ≥1 dose)	No KM curve 28 (20.8, NE)	No KM curve 8.3 (5.3, 11.6)	35 (42.2%; 95% CI, 31.4-53.5) ≥3L: 29 (40%) Cheson 2007 ³⁷	1 (1%) Cheson 2007 ³⁷	17 (20.5%)
Umbralisib	UNITY-NHL ^{8,39,b} N = 117	Histologically confirmed B-cell indolent NHL RR FL after ≥2 prior therapies ECOG PS 0-2	FL	≥2L 101 (86)	FL ≥2L	FL ≥2L	ITT (all patients who received ≥1 dose of single-agent umbralisib)	NA	KM curve 10.6 (7.2, 13.7)	53 (45.3%; 95% CI, 36.1-54.8) Lugano 2014 ²⁷	6 (5.1%; 95% CI, 1.9-10.8) Lugano 2014 ²⁷	14 (12%)

(continued on next page)

Table 2 (continued)

Comparator	Study, N	Key Inclusion Criteria	Population		Baseline Data Available	Outcome Reporting	Analysis Population	Key Efficacy Outcomes				
			Mixed FL n (%) FL	Line of Therapy N (%) ≥3L Median (Range PL)				OS Median (95% CI)	PFS Median (95% CI)	ORR n (%; 95% CI)	CR n (%; 95% CI)	Dis-continuation Due to AEs
Rituximab and lenalidomide												
Rituximab and lenalidomide	AUGMENT ^{40–42, b} N = 178	RR FL grade 1 to 3a, MZL or MCL ≥1 prior therapy (excluding lenalidomide) Relapsed, refractory or progressive disease and not rituximab refractory ECOG PS 0-2	Mixed 147 (83)	Mixed 76 (43) FL 69 (46.9)	FL ≥2L	FL ≥2L	ITT (all patients randomly assigned, regardless of receiving trial treatment)	KM curve Median not reached	KM curve 39.4 (25.1, NE)	118 (80.3%; (95% CI, 72.9–86.4) Cheson 2007 ³⁷	51 (34.7%; 95% CI, 27.0–43.0) Cheson 2007 ³⁷	12 (8.2%) ^c
Tazemetostat												
Tazemetostat	NCT01897571 ^{13, 43, a} N = 99	RR FL grade 1 to 3b including transformed FL ≥2 prior therapies ECOG PS 0-2	FL	≥3L 96 (97)	FL ≥3L	FL ≥3L	Modified ITT (all patients who received ≥1 dose)	EZH2 mutant: KM curve median not reached EZH2 wild type: KM curve median not reached	EZH2 mutant: KM curve 13.8 (10.7, 22) EZH2 wild type: KM curve 11.1 (3.7, 14.6)	EZH2 mutant: 31 (69%; 95% CI, 53–82) EZH2 wild type: 19 (35%; 95% CI, 23–49) Cheson 2007 ³⁷	EZH2 mutant: 6 (13%) EZH2 wild type: 2 (4%) Cheson 2007 ³⁷	5 (9.26%)

Abbreviations: ≥ 3 L = third-line setting or later; AE = adverse event; CART = chimeric antigen receptor T-cell; CI = confidence interval; CR = complete response; ECOG PS = Eastern Cooperative Oncology Group performance status; FL = follicular lymphoma; IPI = International Prognostic Index; ITT = intention to treat; KM = Kaplan-Meier; MAIC = matching-adjusted indirect comparison; MCL = mantle cell lymphoma; MZL = marginal zone lymphoma; n/a = not applicable; NA = not assessed; NE = not estimable; NHL = non-Hodgkin lymphoma; NR = not reported; ORR = objective response rate; OS = overall survival; PFS = progression-free survival; PL = prior lines; RR = relapsed refractory; SD = stable disease; SLL = small lymphocytic lymphoma.

^a Full paper and conference materials.

^b Full paper.

^c Discontinuation of lenalidomide due to AEs.

Indirect Treatment Comparisons of Mosunetuzumab

Due to the lack of a common comparator arm, balance in all prognostic factors and effect modifiers is required to obtain an unbiased estimation of relative treatment effects. MAIC weights were estimated using a propensity score model via the methods of moments. The weights from the matching process were then used to estimate weighted treatment effects to compare outcomes between balanced treatment arms.

To determine whether certain patients were particularly influential in the analysis, the distribution of weights was visually inspected and truncation was considered for weights that were unambiguously outliers (eventually not required). Percentile and bias corrected and accelerated (BCa) methods were used for bootstrapping CIs for weighted HRs and ORs, with 95% CIs obtained via the percentile method selected for the base-case analysis.

An iterative approach was employed to identify the most appropriate base-case scenario that maximized the bias/variance tradeoff for each comparison (Supplemental Methods).

PSA

The preferred target estimand was the average treatment effect (ATE), where feasible, and 2 different methods were used to minimize between group imbalances before indirectly comparing mosunetuzumab with comparator treatments, as recommended by NICE Technical Support Document 17²¹: 1) inverse probability of treatment weighting (IPTW) and 2) propensity score matching (Supplemental Methods).

IPTW⁴⁴ was used as the preferred adjustment method for the base case as it does not discard any units from the sample. Different matching methods were tested, owing to the limited and relatively comparable sample sizes across treatment groups: nearest-neighbor matching⁴⁵ without and with replacement, optimal pair matching,⁴⁶ genetic matching⁴⁷ and full matching.^{46,48} For matching on the propensity score, full matching was preferred over other methods (if similar covariate balance was achieved), as this is the only matching method that allows direct estimation of the ATE. A logistic regression model was used to estimate the propensity scores. Multiple models were tested, including 2-way interaction terms based on expert medical advice (ie, age and number of prior therapies, age and refractory to last line of therapy and progression of disease within 24 months [POD24] and refractory to last line of therapy), as recommended by NICE.²¹ These exploratory analyses were conducted using IPTW and the matching method that resulted in the highest covariate balance as the methods of reference. The model resulting in the highest covariate balance, as measured by the absolute standardized mean differences (threshold of 0.1)⁴⁹ and the complements of overlapping coefficients,⁵⁰ was selected for the final analyses.

The distribution of estimated propensity scores was inspected to assess the overlap between treatment groups and to identify propensity scores with extreme values. Stabilized weights were used in all IPTW analyses. Weight truncation was considered if some unambiguously large IPTW-ATE weights were still found to persist after stabilization. Weights were used in subsequent outcome models to estimate relative treatment effects. If some key covariates were still imbalanced postadjustment, these were considered for inclusion in the outcome model, following a “doubly robust” approach. If certain

regression coefficients or standard errors in the propensity score or outcome models turned out to be implausibly large, the problematic covariates were identified and excluded from the respective models.

For all outcomes, CIs for relative treatment effects of interest were estimated using bootstrapping, as cluster-robust standard errors can be biased for ORs and should be used with caution.^{22,51} For consistency, bootstrapping was used for both binary and survival outcomes, as this method has been shown to be valid in several simulation studies.^{22,52} The percentile method was used to estimate 95% CIs.

Results

SLR and MAIC Feasibility Results

A total of 172 studies described in 214 publications were identified in the SLR (Figure 1, Supplemental Figure 1). Of these, 42 studies evaluated key interventions of interest (Figure 2) and were initially considered as eligible to be included in the MAIC feasibility assessment.^{6-8,13,14,23,33,40,53-58} Figure 2 summarizes how these studies were further filtered before an in-depth MAIC feasibility assessment was conducted. Twenty-one studies (Supplemental Table 6) were excluded because they did not meet the selection criteria. The remaining 21 studies were further categorized using the criteria outlined previously (Supplemental Table 7). Where multiple studies for a given comparator treatment were deemed feasible for conducting a MAIC, only the most appropriate study was prioritized and the others were excluded (Supplemental Table 8).

An overview of the analysis populations and key outcomes reported in the comparator studies found to be the most suitable for MAIC analyses is given in Table 2. Five studies were considered for ITC by PSA (Table 3).

ITC Results

The key methodological aspects, results and the effective sample size (ESS; a measure of the sample size postadjustment and of robustness) of each MAIC analysis are described in Table 4 and in the Supplemental Methods and Results. Figure 3 shows a summary overview of the main ITC findings. Evidence was classified as relatively weakly or strongly in favor of a comparator, depending on whether the CIs around point estimates did or did not cross 1 in both the base-case and sensitivity analyses. This was done also to take into consideration potential differences in study design that could not be resolved via these methods.

CART Therapies

The axicabtagene ciloleucel MAIC analyses were conducted based on 65 (72%) mosunetuzumab-treated patients in GO29781 who did not have stable disease (defined as patients who relapsed within 1 year from completion of last therapy) to align with the same exclusion criteria used in ZUMA-5 (Table 4).²³⁻²⁶

The base-case scenario maximized the bias/variance tradeoff while controlling for all feasible priority prognostic factors. A sensitivity analysis, termed scenario 1, explored the impact of controlling for age as a median rather than a mean (see Supplemental Table 9 and Supplemental Figure 2 for a comparison of baseline characteristics

Figure 1 Flow diagram of study interventions in studies identified in the clinical SLR. Key interventions of interest for the MAIC feasibility assessment are indicated in bold. Some boxes include more comparators than the number of studies because some studies were comparative. ASCT = autologous stem cell transplant; FL = follicular lymphoma; MAIC = matching-adjusted indirect comparison; PL = prior lines of therapy; R-CHOP = rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone; SCT = stem cell transplant; SLR = systematic literature review.

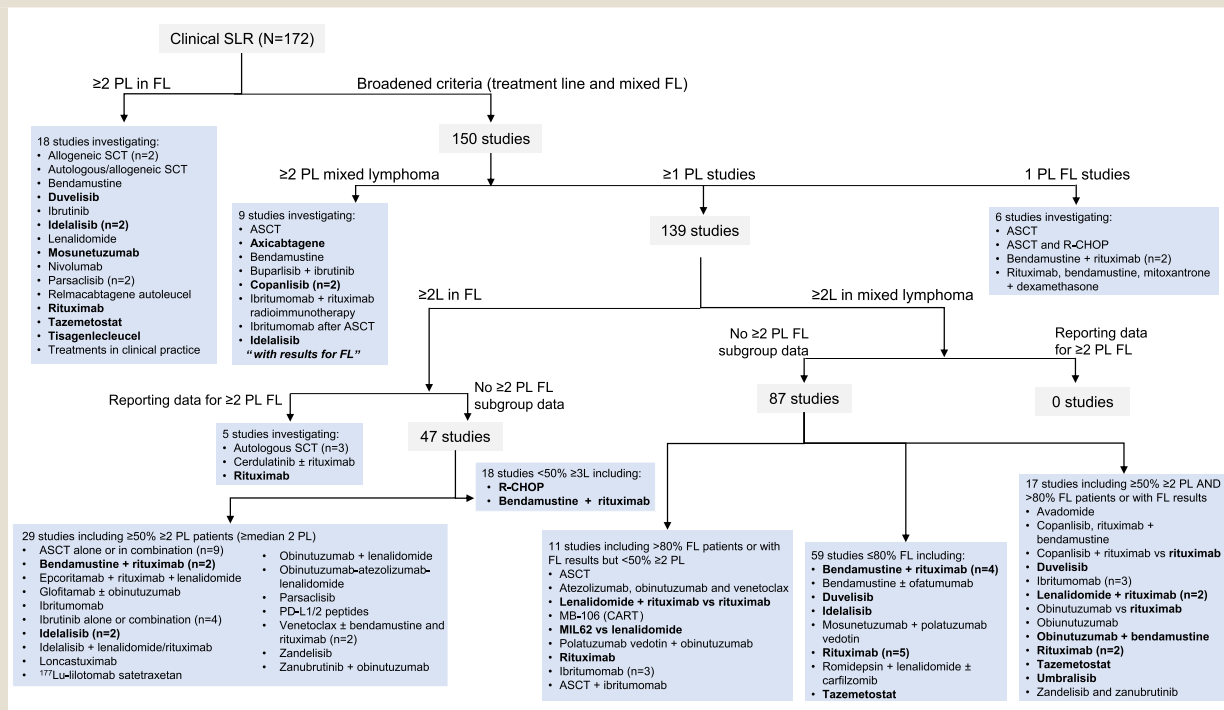
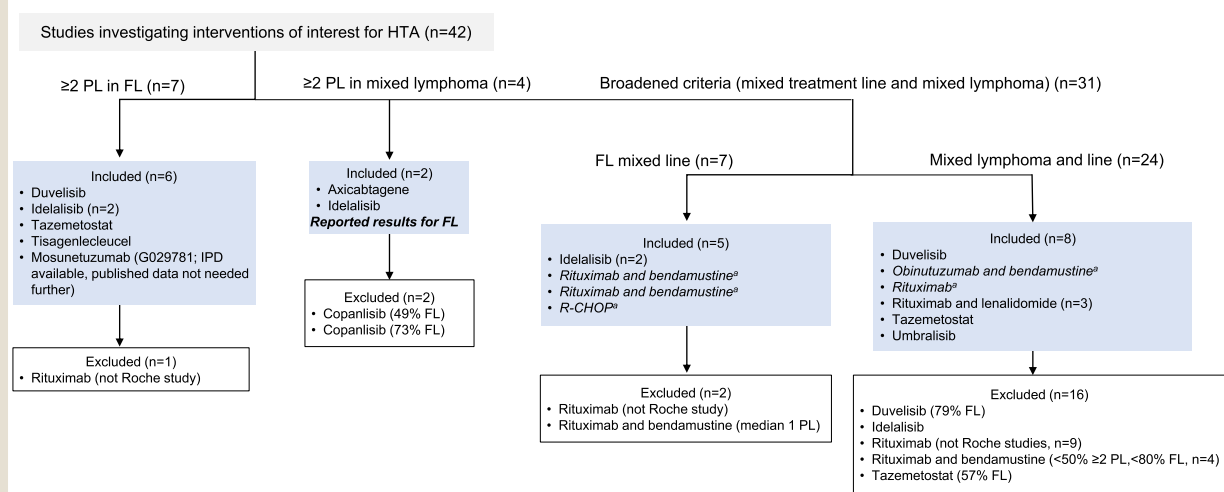


Figure 2 Flow diagram of comparator studies available for inclusion in the MAIC feasibility assessment. FL = follicular lymphoma; HTA = health technology assessment; IPD = individual patient-level data; MAIC = matching-adjusted indirect comparison; PL = prior lines of therapy; PSA = propensity score analysis. ^aStudy is Roche sponsored, and IPD are available. It was not considered further in the MAIC feasibility assessment but was considered for PSAs.



Indirect Treatment Comparisons of Mosunetuzumab

Figure 3 Summary overview of ITC results. AE = adverse event; CR = complete response; ITC = indirect treatment comparison; ORR = objective response rate; OS = overall survival; PFS = progression-free survival; WT = wild type. ^aEvidence was classified as relatively strongly or weakly in favor of a comparator depending on whether confidence intervals around point estimates did or did not cross 1 in both the base-case and sensitivity analyses. ^bResults are biased in favor of the comparator due to substantial differences in enrollment criteria from GO29781 and should be interpreted with a high degree of caution, particularly regarding OS and PFS.

Mosunetuzumab versus: (STUDY)	CAR-T therapy		PI3K inhibitors			Anti-CD20-containing regimens			
	Axicabtagene ciloleucel (ZUMA-5)	Tisagenlecleucel (ELARA)	Idelalisib (DELTA)	Duvelisib (DYNAMO)	Umbralisib (UNITY-NHL)	Tazemetostat (NCT01897571, WT population)	Rituximab + lenalidomide ^b (AUGMENT)	Obinutuzumab + bendamustine (GADOLIN)	Rituximab + bendamustine (CONTRALTO + GO29365)
Response assessment criteria used	✗	✗	✓	✓	✗	✓	✓	✓	✗
OS	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Base case point estimate very close to 1	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Not available or not applicable	Not available or not applicable	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring comparator ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a
PFS	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Not available or not applicable	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring comparator ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a
ORR	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a
CR	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	1-2 events in comparator trial	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	1-2 events in comparator trial	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a
Tolerability (discontinuation due to AEs)	Not available or not applicable	Not available or not applicable	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	Relatively strong / weak evidence of difference favoring mosunetuzumab ^a	1-2 events in comparator trial

for the base-case and scenario analyses and diagnostic plots, respectively).

The evidence from the ITC suggested a numerical trend in favor of mosunetuzumab compared with axicabtagene ciloleucel for the OS HR, although the uncertainty around the point estimate was relatively large (Table 4, Supplemental Table 10, Supplemental Figure 3A). The PFS HR and CR OR (Supplemental Tables 10 and 11 and Supplemental Figure 3B) were strongly in favor of axicabtagene ciloleucel compared with mosunetuzumab, in that the CI around the point estimate did not cross 1 (Table 4). Conversely, while the base-case ORR results more weakly favored axicabtagene ciloleucel compared with mosunetuzumab, because the CI around the point estimate crossed 1, this conclusion was not supported by the scenario 1 analysis (Supplemental Table 11).

The tisagenlecleucel MAIC analyses were performed based on 78 (87%) mosunetuzumab-treated patients who 1) were either refractory to a 2L or later line of systemic therapy or relapsed within 6 months after completion of a 2L or later line of systemic therapy; 2) relapsed during anti-CD20 antibody maintenance (following ≥ 2 lines of therapy) or within 6 months after maintenance therapy completion; or 3) relapsed after autologous hematopoietic stem-cell transplant (ASCT), to align with the same exclusion criteria used in ELARA (Table 4).^{12,28–32} The base-case analysis maximized the bias/variance tradeoff while controlling for all high-priority prognostic factors that were feasible. An additional scenario 1 analysis explored the impact of excluding double refractoriness (whose

actual prognostic value seems to be unclear) and Eastern Cooperative Oncology Group (ECOG) performance status (PS) (1 vs. 0), as well as of controlling for age as a median (see Supplemental Table 12 and Supplemental Table 4 for a comparison of baseline characteristics for the base-case and scenario analyses and diagnostic plots, respectively).

ITC results with respect to OS, ORR, and CR weakly favored tisagenlecleucel with a numerical trend in the point estimate (Table 4, Supplemental Tables 13 and 14, Supplemental Figure 5A), which was, however, associated with a CI crossing 1. The PFS HR and CI from the ITCs were strongly in favor of tisagenlecleucel compared with mosunetuzumab (Table 4, Supplemental Table 13, Supplemental Figure 5B).

PI3K Inhibitors

All base-case scenarios versus each individual PI3K inhibitor analyzed maximized the ESS while at the same time controlling for all high-priority prognostic factors that were feasible.

The analyses for the idelalisib MAIC were based on 48 (53%) GO29781 patients who were refractory to both a rituximab- and an alkylating agent-containing regimen, to align with the same inclusion criteria used in DELTA (Table 4).^{6,33–35}

The base-case analysis used values from Flatiron to inform the proportion of patients with low hemoglobin (Hgb) levels and excluded the covariates of time since completion of last therapy

Table 3 Overview of Comparator Studies Selected for PSA

Comparator	Study N in Comparator Arm Studied	Key Study Inclusion Criteria	Filtered Population Selected for PSA	Key Efficacy Outcomes (KM Curves and Response Criteria Indicated if Available)			
				OS Median (95% CI)	PFS Median (95% CI)	ORR (%)	CR (%)
Obinutuzumab plus bendamustine	GADOLIN ^{54,a} (n = 194)	iNHL including grade 1-3a FL ECOG PS 0-2 Refractory to rituximab	≥3L FL, excluded ECOG PS 2 n = 77	KM curve NE (NE-NE)	KM curve IRC assessed: NR (22.5-NR) after follow-up of 21.9 (IQR 12.1-31.0) mo	79% (by IRC after 12 mo)	17% (by IRC after 12 mo)
Rituximab plus bendamustine	CONTRALTO ^{53,a} (n = 50)	RR FL grade 1-3a ≥1 prior therapy	≥3L RR FL, excluded ECOG PS 2 n = 46	KM curve (median not reported)	KM curve (median not stated)	57% by INV-assessed PET at 12 mo	51% by INV-assessed PET at 12 mo
	GO29365 ^{56,b} (n = 41)	RR FL and DLBCL Transplant ineligible		Not reported in abstract	17.3 (12.5, NR)		63% by IRC-assessed PET 6-8 weeks after treatment end

Abbreviations: CI = confidence interval; CR = complete response; DLBCL = diffuse large B-cell lymphoma; ECOG PS = Eastern Cooperative Oncology Group performance status; FL = follicular lymphoma; IRC = independent review committee; iNHL = indolent non-Hodgkin lymphoma; INV = investigator; KM = Kaplan-Meier; NE = not estimable; NR = not reached; ORR = objective response rate; OS = overall survival; PET = positron emission tomography; PFS = progression-free survival; PSA = propensity score analysis; RR = relapsed refractory.

^a Full paper.

^b Abstract and presentation.

and ECOG PS (1 vs. 0). An additional scenario 1 analysis explored the impact of controlling for all reported prognostic factors and of assuming the same value for the proportion of patients with low Hgb levels as in the GO29781 trial (see Supplemental Table 15 and Supplemental Figure 6 for a comparison of baseline characteristics in the base-case and scenario analyses and diagnostic plots, respectively).

Evidence from the ITC with respect to OS and ORR overall weakly favored mosunetuzumab over idelalisib only with a numerical trend (Table 4, Supplemental Tables 16 and 17, Supplemental Figure 7A), but the uncertainty around the point estimates was large and the CIs crossed 1. With regard to PFS, the ITC results weakly favored idelalisib over mosunetuzumab with a numerical trend, and large uncertainty around the point estimate remained (Table 4, Supplemental Table 16, Supplemental Figure 7B). The ORs and CIs for CR and treatment discontinuation due to AEs were generally strongly in favor of mosunetuzumab compared with idelalisib (Table 4, Supplemental Tables 17 and 18). However, the results for discontinuation due to AEs were not supported by the BCa method for estimating the bootstrapped CIs.

The MAIC analyses for duvelisib were conducted in 71 (79%) GO29781 patients who were refractory to prior anti-CD20-containing therapy, to better align with the inclusion criteria used in DYNAMO (Table 4).^{7,38} The base-case analysis excluded the covariates of ECOG PS (1 vs. 0), time since completion of last therapy and double refractoriness to both an anti-CD20 therapy and an alkylating agent, while controlling for age as a median. As the ESS resulting from all other exploratory analyses (not shown) was unacceptably small (up to ≈20%, corresponding to <20 patients), only the base-case and unadjusted analysis results are presented (see Supplemental Table 19 and Supplemental Figure 8 for a comparison of baseline characteristics in the base-case analysis and diagnostic plots, respectively).

Analyses for OS and PFS were not feasible as Kaplan-Meier curves were not reported for DYNAMO. The ORs and CIs for ORR and CR overall strongly favored mosunetuzumab versus duvelisib, as they did for discontinuation due to AEs (Table 4, Supplemental Table 20).

For the umbralisib MAIC, harmonizing the inclusion and exclusion criteria between GO29781 and UNITY-NHL^{8,39} was not feasible, mainly due to 13.7% and 3.4% of the patients in UNITY-NHL having 2L treatment and ECOG PS ≥2, respectively, compared with 0% for both variables in GO29781. Hence, all analyses were based on the full pivotal cohort of 90 patients with FL in GO29781 who had received ≥2 prior therapies. Different approaches to control for the proportion of patients with prior ASCT were explored as this prognostic factor was defined differently in the 2 studies, mainly due to different inclusion criteria. The base case controlled for age as a mean; excluded ECOG PS, presence of B symptoms and time since completion of last therapy; and assumed the proportion of patients with prior ASCT to be the same as in the GO29781 trial. Two additional scenarios explored the impact of controlling for all reported prognostic factors and the impact of excluding prior ASCT in addition to ECOG PS, presence of B symptoms and time since completion of last therapy, as this resulted in the highest ESS percentage among all sensitivity analyses (see

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Table 4 Summary of Results of Base-Case Analyses

Comparator (source of data and any limitations)	Mosunetuzumab cohort details and ESS after matching (absolute numbers)	Covariates used in base case analysis		Base-case MAIC results for mosunetuzumab versus comparator				
			Method for estimating HR/OR	OS (HR)	PFS (HR)	ORR (OR)	CR (OR)	Discontinuation due to AEs (OR)
CART therapy								
Axicabtagene ciloleucel ZUMA-5 (n=86) ≥3L RR FL	Analyses conducted in patients without stable disease (defined as patients who relapsed within 1 year from completion of last therapy) to align with the same exclusion criteria used in ZUMA-5 (n=65) ESS=37.7	Age, mean ECOG (1 vs 0), % FLIPI ≥3 (yes), % Ann Arbor Stage IV (yes), % Size of the largest node lesion (cm), mean No. of prior therapies ≥3, % Refractory to last line, % POD24, % Prior ASCT, % Bone marrow involvement, % Time since completion of last therapy (months), mean	HR/OR (95% CI) from unadjusted Cox/logistic regression model	0.875 (0.351, 2.179)	2.365 (1.425, 3.924)	0.225 (0.076, 0.662)	0.309 (0.151, 0.630)	Not feasible
			Bootstrap median HR/OR (95% percentile CI) from weighted Cox/logistic regression model	0.556 (0.025, 1.466)	2.205 (1.339, 3.244)	0.300 (0.147, 1.212)	0.336 (0.179, 0.635)	Not feasible
Tisagenlecleucel ELARA (n=94) ≥2L RR FL	Analyses were conducted in patients who were either: 1) refractory to ≥2L systemic therapy or relapsed within 6 months after completion of ≥2L systemic therapy; 2) relapsed during anti- CD20 antibody maintenance (following ≥2 lines of therapies as above) or within 6 months after maintenance completion; 3) relapsed after autologous HSCT (n=78) ESS=39.5	Age, mean ECOG (1 vs 0), % FLIPI ≥3, % Ann Arbor Stage III–IV, % No. of prior therapies ≥4, % Refractory to last line, % Refractory to prior anti- CD20 regimen, % Double refractory, % High LDH, % POD24, % Bulky disease, % Bone marrow involvement, % Prior ASCT, %	HR/OR (95% CI) from unadjusted Cox/logistic regression model	1.104 (0.426, 2.863)	1.600 (1.025, 2.497)	0.576 (0.260, 1.275)	0.577 (0.309, 1.080)	Not feasible
			Bootstrap median HR/OR (95% percentile CI) from weighted Cox/logistic regression model	1.049 (0.157, 2.458)	1.784 (1.032, 2.629)	0.641 (0.306, 1.661)	0.619 (0.298, 1.268)	Not feasible
P13K inhibitors								
Idelalisib DELTA (n=72) ≥3L RR FL (subgroup)	Analyses were conducted in patients who were double refractory to both a rituximab- and an alkylating agent- containing regimen to align with the same inclusion criteria used in DELTA (N=48) ESS=29.8	Age, mean FLIPI ≥3, % Ann Arbor Stage III–IV, % Size of the largest node lesion (cm), mean No. of prior therapies ≥4, % Refractory to last line, % High LDH, % Low Hgb level, % (imputed for DELTA) POD24, % Prior ASCT, %	HR/OR (95% CI) from unadjusted Cox/logistic regression model	0.660 (0.290, 1.503)	1.112 (0.682, 1.815)	1.943 (0.893, 4.225)	5.000 (2.160, 11.573)	0.216 (0.059, 0.783)
			Bootstrap median HR/OR (95% percentile CI) from weighted Cox/logistic regression model	0.823 (0.087, 1.586)	1.247 (0.657, 2.167)	1.562 (0.665, 4.833)	4.561 (2.127, 10.167)	0.355 (0.000, 0.941) ^a

(continued on next page)

Table 4 (continued)

Duvelisib DYNAMO (n=83) ≥2L RR FL (subgroup with 83% of patients receiving ≥3L treatment)	Analyses conducted in patients who were refractory to prior rituximab containing therapy to align with the same inclusion criteria used in DYNAMO (n=71) ESS≈28.6	Age, median FLIPI ≥3, % Ann Arbor Stage III–IV, % No. of prior therapies ≥3, % Refractory to last line, % High LDH, % POD24, % Bulky disease, %	HR/OR (95% CI) from unadjusted Cox/logistic regression model	Not feasible	Not feasible	4.714 (2.325, 9.559)	99.937 (13.171, 758.300)	0.171 (0.048, 0.612)
			Bootstrap median HR/OR (95% percentile CI) from weighted Cox/logistic regression model	Not feasible	Not feasible	4.598 (1.893, 15.135)	104.426 (48.397, 225.868)	0.047 (0.000,0.310) ^a
Umbralisib UNITY-NHL (n=117) ≥2L RR FL (86% receiving ≥3L)	Analyses were conducted on the full ≥3L FL pivotal cohort from GO29781 (n=90) ESS≈24.9	Age, mean Ann Arbor Stage III–IV, % No. of prior therapies ≥3, % Refractory to last line, % Refractory to prior aCD20 regimen, % High LDH, % Bulky disease, % Prior ASCT, %	HR/OR (95% CI) from unadjusted Cox/logistic regression model	Not feasible	0.654 (0.450, 0.950)	4.830 (2.568, 9.086)	27.750 (11.022, 69.869)	0.342 (0.109, 1.078)
			Bootstrap median HR/OR (95% percentile CI) from weighted Cox/logistic regression model	Not feasible	0.329 (0.166, 0.608)	8.207 (3.649, 122.105)	50.303 (22.987, 154.931)	0.039 (0.000,0.237) ^a
Rituximab and lenalidomide								
Rituximab plus lenalidomide AUGMENT (N=147) ≥2L mixed lymphoma (FL subgroup [47% received ≥2 prior therapies], 100% of patients not refractory to prior rituximab)	All analyses are based on the full ≥3L FL pivotal cohort from GO29781 (n=90) ESS≈32.9	Age, mean FLIPI ≥3, % Ann Arbor Stage III–IV, % Refractory to last line, % High LDH, % Low Hgb level, % (imputed for AUGMENT) Bone marrow involvement, % POD24, % Bulky disease, %	HR/OR (95% CI) from unadjusted Cox/logistic regression model	2.002 (0.739, 5.428)	2.223 (1.463, 3.378)	0.983 (0.510, 1.897)	2.824 (1.643, 4.852)	0.523 (0.163, 1.675)
			Bootstrap median HR/OR (95% percentile CI) from weighted Cox/logistic regression model	1.419 (0.135, 3.371)	1.449 (0.648, 2.872)	1.170 (0.549, 4.589)	5.018 (2.516, 11.195)	0.585 (0.001, 1.606)
Tazemetostat								
Tazemetostat NCT01897571 (n=54) ≥3L RR FL (without EHZ2 mutation) (included patients with Grade 3b and trFL)	Analyses were conducted in patients without an EHZ2 mutation to align with Morschhauser 2020 WT cohort (n=82) ESS≈36.9	Age ≥61 years, % ECOG (1 vs 0), % No. of prior therapies ≥3, % Refractory to last line, % Refractory to prior anti- CD20 regimen, % Double refractory, % POD24 (yes), % Bulky disease, % Prior ASCT, %	HR/OR (95% CI) from unadjusted Cox/logistic regression model	0.318 (0.129, 0.781)	0.571 (0.354, 0.921)	7.599 (3.479, 16.596)	42.774 (9.726, 188.123)	0.503 (0.129, 1.963)

(continued on next page)

Supplemental Table 21 and Supplemental Figure 9 for a comparison of baseline characteristics in the base-case and scenario analyses and diagnostic plots, respectively).

An analysis for OS was not feasible as a Kaplan-Meier curve was not reported for UNITY-NHL. The HR for PFS and the ORs for ORR and CR, as well as their respective CIs, overall strongly favored mosunetuzumab versus umbralisib in the base-case (Table 4) and

sensitivity analyses (Supplemental Tables 22 and 23, Supplemental Figure 10).

The ORs and percentile bootstrapped CIs for discontinuation due to AEs (Supplemental Table 24) strongly favored mosunetuzumab versus umbralisib in both scenario analyses and were concordant with the conclusions from the base-case scenario (Table 4). However, the ORs and BCa bootstrapped CIs (Supple-

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Table 4 (continued)

Comparator (source of data)	Filtering process to align patient populations prior to indirect comparison	Covariates used in analyses	PSA results for the comparison of mosunetuzumab versus comparator					
			Method for estimating HR/OR	OS (HR)	PFS (HR)	ORR (OR)	CR (OR)	Discontinuation due to AEs (OR)
Obinutuzumab plus bendamustine GADOLIN ³⁶ Cohort of ≥3L FL patients	Patients with ECOG PS 2 excluded from the GADOLIN cohort (to align with the inclusion/exclusion criteria of the GO29781 trial) and patients who were not refractory to a prior rituximab-containing regimen excluded from the GO29781 cohort (to align with the inclusion/exclusion criteria of the GADOLIN trial) Mosunetuzumab (n=71) Obinutuzumab plus bendamustine (n=77)	Age, mean ECOG PS (1 vs 0), % FLIPI ≥3, % Ann Arbor Stage III/IV, % No. of prior therapies ≥3, % Refractory to last line, % POD24, % Prior ASCT, % Size of largest node lesion (cm), mean Double refractory, % Low Hgb, % High LDH, % Bone marrow involvement, % Time since completion of last therapy (months), mean Presence of B symptoms, %	Unadjusted	0.60 (0.26, 1.39)	2.06 (1.27, 3.32)	1.05 (0.49, 2.26)	4.65 (2.26, 9.57)	0.30 (0.08, 1.12)
			Full matching plus covariate adjustment	0.73 (0.18, 1.85)	2.01 (1.20, 5.10)	1.04 (0.40, 4.32)	5.85 (2.60, 24.37)	0.25 (0.02, 0.93)
			IPTW	0.71 (0.23, 1.75)	2.25 (1.29, 4.16)	1.30 (0.48, 3.57)	4.66 (1.76, 12.70)	0.23 (0.04, 0.92)
Rituximab plus bendamustine CONTRALTO and G029365 Cohort of ≥3L FL patients	Patients with ECOG PS 2 excluded from CONTRALTO and GO29365 cohorts (to align with the inclusion/exclusion criteria of the GO29781 trial) and patients who received >5 prior anticancer regimens excluded from the GO29781 cohort (to align with the populations enrolled in CONTRALTO and GO29365) Mosunetuzumab (n=81) Rituximab plus bendamustine (n=46)		Unadjusted					
			(sample size=127)	0.85 (0.27, 2.72)	1.09 (0.66, 1.82)	0.50 (0.17, 1.46)	0.51 (0.23, 1.14)	2.34 (0.25, 21.57)
			Optimal pair matching plus covariate adjustment	0.44 (0.03, 3.20)	0.90 (0.38, 1.74)	0.65 (0.12, 9.43)	0.56 (0.24, 2.16)	3.05 (0.09, 7.84)
			(sample size=92)					
			IPTW plus covariate adjustment	0.49 (0.08, 2.30)	0.85 (0.48, 1.46)	0.73 (0.14, 2.18)	0.58 (0.22, 1.35)	1.76 (0.20, 5.37)

Abbreviations: ≥2L = second-line therapy or beyond; ≥3L = third-line therapy or beyond; AE = adverse event; ASCT = autologous stem cell transplant; BCa = bias corrected and accelerated; BR = bendamustine plus rituximab; CI = confidence interval; CR = complete response; ECOG PS = Eastern Cooperative Oncology Group performance status; ESS = effective sample size; FL = follicular lymphoma; FLIPI = Follicular Lymphoma International Prognostic Index; Hgb = hemoglobin; HR = hazard ratio; HSCT = hematopoietic stem cell transplant; IPD = individual patient-level data; IPTW = inverse probability of treatment weighting; LDH = lactate dehydrogenase; M = mosunetuzumab; MAIC = matching-adjusted indirect comparison; OB = obinutuzumab plus bendamustine; OR = odds ratio; ORR = overall response rate; OS = overall survival; PFS = progression-free survival; POD24 = progression of disease within 24 months of start of initial therapy; PSA = propensity score analysis; RR = relapsed refractory; trFL = transformed follicular lymphoma; WT = wild type.

Results where the CI did not cross 1 are shown in bold text. Point estimates favoring mosunetuzumab are indicated in green (darker green indicates results where the CI did not cross 1) while those favoring the comparator are indicated in blue (darker blue indicates results where the CI did not cross 1). OR > 1 is in favor of mosunetuzumab regarding ORR and CR but against mosunetuzumab regarding discontinuations, while HR ≤ 1 is in favor of mosunetuzumab regarding PFS and OS.

^a Results not supported by the BCa method for the estimation of the bootstrapped CIs.

mental Table 24) for the base-case and scenario 2 analyses only weakly favored mosunetuzumab, as the OR CIs crossed 1. This suggests that the bootstrapped sample of ORs for discontinuation due to AEs might not be normally distributed, and thus the CI estimates obtained from the percentile method should be interpreted with some degree of caution.

Rituximab and Lenalidomide

For the MAIC with rituximab plus lenalidomide, harmonizing the inclusion/exclusion criteria between GO29781 and AUGMENT^{40–42} was not feasible. This was mainly because in the AUGMENT FL subgroup, ≈53% of patients received only 1 prior treatment and 0% were refractory to rituximab, compared with

0% and $\approx 79\%$ in the GO29781 trial, respectively. Attempts were made to conduct the comparison in the subgroup of patients who were refractory to an anti-CD20 antibody-containing regimen in the GO29781 trial. However, this resulted in a very small sample size after filtering and before adjustment ($n = 19$), which did not allow us to control for all key prognostic factors due to an unacceptably small ESS ($<10\%$ of the original sample size) and thus was not pursued further. Therefore, all analyses were based on the full cohort of 90 patients with RR FL in GO29781 who received ≥ 2 prior therapies and 147 patients in AUGMENT who had FL and received ≥ 1 line of prior therapy. The imbalance in the number of prior lines of therapy and in the refractory status to prior anti-CD20 treatment introduced an important bias against mosunetuzumab in all analyses.

The base-case analysis used the same value for the proportion of patients with low Hgb levels as in GO29781 and (in addition to prior therapy lines and refractoriness to rituximab, which could not be controlled for) excluded ECOG PS, presence of B symptoms and time since completion of last therapy, as this approach maximized the bias/variance tradeoff while controlling for all feasible high-priority prognostic factors. An additional scenario 1 analysis explored the impact of controlling for all reported prognostic factors that were feasible and using Flatiron as the data source to inform the value for the proportion of patients with low Hgb levels (see Supplemental Table 25 and Supplemental Figure 11 for a comparison of baseline characteristics in the base-case and scenario analyses and diagnostic plots, respectively).

The ITC of mosunetuzumab versus rituximab plus lenalidomide showed mixed results. With regard to OS and PFS, evidence overall weakly favored rituximab plus lenalidomide, with a numerical trend in favor of this comparator but large uncertainty around the point estimates and CIs crossing 1 (Table 4, Supplemental Table 26, Supplemental Figure 12). The ORs and CIs for CR strongly favored mosunetuzumab versus rituximab plus lenalidomide, but ORs for ORR only showed a numerical trend favoring mosunetuzumab in the base-case and not the scenario 1 analysis, with large uncertainties and CIs crossing 1 in all analyses (Table 4, Supplemental Table 27). The OR for treatment discontinuation due to AEs only weakly favored mosunetuzumab, with a numerical trend in favor of this comparator but relatively large uncertainty around all point estimates and CIs crossing 1 (Table 4, Supplemental Table 27).

Tazemetostat

The tazemetostat MAIC was conducted based on patients who did not have an *EZH2* mutation, to align with the *EZH2*-wild-type cohort in NCT01897571.¹³ Because only 8 patients in GO29781 had this mutation, it was not possible to perform the comparisons in the *EZH2*-mutant population nor to control for this factor, which is a clear effect modifier for the *EZH2* inhibitor tazemetostat. The base case maximized the bias/variance tradeoff while controlling for all feasible priority prognostic factors and for age as a median. An additional scenario 1 analysis explored the impact of controlling for age as a mean and not controlling for double refractoriness (see Supplemental Table 28 and Supplemental Figure 13 for a comparison of baseline characteristics in the base-case and scenario analyses and diagnostic plots, respectively).

In patients without *EZH2* mutations, the point estimates and respective CIs for OS, PFS, ORR, CR and discontinuation due to AEs strongly favored mosunetuzumab compared with tazemetostat in almost all analyses conducted (Table 4, Supplemental Tables 29 and 30, Supplemental Figure 14). However, the ORs and BCa bootstrapped CIs for the base-case and scenario 1 analyses for discontinuation due to AEs only weakly favored mosunetuzumab, as the OR CIs crossed 1, and were thus not in strong agreement with the results of the percentile bootstrap method. This suggests that the bootstrapped sample of ORs for discontinuation due to AEs might not be normally distributed; hence the CI estimates obtained from the percentile method should be interpreted with some degree of caution.

Obinutuzumab Plus Bendamustine

The population used for this indirect comparison was a cohort of 80 patients with FL who had received ≥ 2 prior therapies in the obinutuzumab plus bendamustine arm of the GADOLIN trial.⁵⁴ To ensure that the patient cohorts used for the analyses were as homogeneous as possible before any indirect comparisons were attempted, patients with ECOG PS 2 were excluded from the GADOLIN cohort, as were those who were not refractory to a prior anti-CD20 antibody-containing regimen from the GO29781 cohort (Table 3), to align with the eligibility criteria of the GO29781 and GADOLIN trials, respectively. This resulted in 71 patients in the mosunetuzumab arm and 77 patients in the obinutuzumab plus bendamustine arm (Table 4). Further methodological details with regard to propensity score model parameters and diagnostics, covariate balance and exploratory analyses conducted for the selection of the final matching method of reference, as well as individual results, are described in the Supplemental Results, Supplemental Table 31 and Supplemental Figures 15-18.

Evidence from the ITC regarding OS and ORR (Table 4, Supplemental Figure 19) overall weakly favored mosunetuzumab over obinutuzumab plus bendamustine with a numerical trend, but the uncertainty around the point estimates was relatively large (particularly for ORR) and the CIs crossed 1 in all analyses. Evidence with respect to ORs and CIs for CR and treatment discontinuation due to AEs was overall strongly in favor of mosunetuzumab over obinutuzumab plus bendamustine, with CIs not crossing 1 in any of the adjusted analyses (Table 4). By contrast, the evidence with respect to PFS HRs and CIs was overall strongly in favor of obinutuzumab plus bendamustine over mosunetuzumab (Table 4, Supplemental Figure 20).

Rituximab Plus Bendamustine

The population used for indirectly comparing mosunetuzumab to rituximab plus bendamustine was combined from 2 cohorts of patients with FL who had received ≥ 2 prior therapies in the rituximab plus bendamustine arms of the CONTRALTO⁵⁸ and the GO29365 (NCT02257567)⁵⁶ trials (Table 3). To ensure that the patient cohorts used for the analyses were as homogeneous as possible before any indirect comparisons were attempted, patients with ECOG PS 2 were excluded from both comparator trial cohorts (to align with the inclusion/exclusion criteria of the GO29781 trial) and patients who received >5 prior anticancer regimens were

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excluded from the GO29781 cohort (to align with the patients enrolled in the comparator cohorts). This resulted in 81 patients in the mosunetuzumab arm and 46 patients in the rituximab plus bendamustine arm (Table 4). Further methodological details with regard to propensity score model parameters and diagnostics, covariate balance and exploratory analyses conducted for the selection of the final matching method of reference, as well as individual results, are described in the Supplemental Results, Supplemental Table 32 and Supplemental Figures 21-24.

The ITC of mosunetuzumab versus rituximab plus bendamustine showed mixed results with respect to the efficacy and tolerability outcomes compared (Table 4). OS and PFS HRs and CIs overall weakly favored mosunetuzumab over rituximab plus bendamustine with a numerical trend (Table 4, Supplemental Figures 25 and 26). ORs and CIs for ORR, CR and discontinuation due to AEs overall weakly favored rituximab plus bendamustine over mosunetuzumab with a numerical trend. Overall, the uncertainty around the point estimates for all endpoints was relatively large (particularly for ORR and most notably discontinuation due to AEs), and the CIs crossed 1 in all analyses (Table 4). Notably, only 1 event of discontinuation due to AEs was observed in the rituximab plus bendamustine cohort used for the comparison; hence the ITC results on tolerability from this comparison should be interpreted with great caution.

Reliable ITCs of mosunetuzumab versus R-CHOP and rituximab monotherapy based on the EORTC 20981⁵⁷ and GAUSS⁵⁵ trials, respectively, were not feasible because of limitations associated with small sample sizes, unavailability of information on certain key prognostic factors and important residual imbalances after adjusting for differences in the available factors.

Discussion

To date, despite the availability of several treatment options, no consensus standard of care exists for RR FL patients who have received ≥ 2 prior therapies. Although randomized controlled trials still remain the most reliable source of evidence, ITCs are helpful in comparing the relative benefits of multiple treatment options, particularly those investigated only in single-arm trials. MAICs and PSAs represent indirect treatment comparison methodologies that are widely used to generate comparative evidence and are increasingly being accepted by local and international agencies in the assessment of new hematological oncology treatments. Beyond efficacy outcomes, tolerability outcomes were preferred over general safety outcomes for the ITCs, as their analysis is associated with fewer limitations.

The ITCs of mosunetuzumab versus CART therapies generally favored axicabtagene ciloleucel relatively strongly regarding most efficacy outcomes and, to a lesser extent, tisagenlecleucel over mosunetuzumab, although important caveats and uncertainties remain. Tolerability, indicated by discontinuation due to AEs, could not be compared because CART therapies are administered as one-off treatments and patients cannot discontinue due to AEs. Although CART therapies have distinct safety profiles, additional safety differences versus mosunetuzumab were not analyzed in this study.

The ITCs of mosunetuzumab versus PI3K inhibitors generally favored mosunetuzumab over duvelisib, umbralisib and idelalisib,

with the strongest evidence in favor of mosunetuzumab seen for CR and ORR rates and, to a lesser extent, tolerability, depending on the specific comparator. Mosunetuzumab was associated with a lower likelihood of treatment discontinuation due to AEs versus PI3K inhibitors, consistent with reports of serious toxicities associated with duvelisib and idelalisib⁵⁹. In patients without the *EZH2* mutation, the results of the ITC strongly favored mosunetuzumab over tazemetostat for all efficacy and, to a lesser extent, for the tolerability outcomes compared.

ITCs versus the anti-CD20 antibody-based regimens yielded mixed results, making it more difficult to draw firm conclusions from the outcome of these comparisons. For the comparison between obinutuzumab plus bendamustine and mosunetuzumab, the CR comparison favored mosunetuzumab, while the PFS comparison favored obinutuzumab plus bendamustine. A relatively higher rate of censoring for PFS was observed in the mosunetuzumab group compared with the obinutuzumab plus bendamustine group, most likely due to considerable differences in the follow-up time between GADOLIN and GO29781. As GO29781 accrues longer follow-up, further investigation may help to clarify the impact of different durations of follow-up on the ITC results.

A recently published MAIC of ZUMA-5 and ELARA suggested that tisagenlecleucel and axicabtagene ciloleucel had comparable ORR, CR, OS and PFS and that tisagenlecleucel had better safety outcomes.⁶⁰ Discrepancies between our results and those of that study are likely attributed to different methodologies used with respect to covariate and patient selection. Another MAIC analysis of tazemetostat versus idelalisib, duvelisib, copanlisib, and umbralisib showed that the *EZH2* inhibitor may have a better safety profile than the PI3K inhibitors, while achieving similar efficacy outcomes.⁵⁹ Taken together, the results of our MAICs versus tazemetostat and PI3K inhibitors suggested that mosunetuzumab could have better tolerability and improved ORRs and CRs compared with duvelisib, idelalisib, umbralisib and tazemetostat.

The findings from this study should be interpreted in the context of their limitations, most of which are inherent in unanchored MAICs: differences in study designs and the rarity and indolent nature of FL in the 3L or later. In many comparisons, there were misalignments in eligibility criteria between studies. Filtering procedures were applied across cohorts whenever possible to improve population overlap before the ITCs were conducted. However, ECOG PS in GO29781 was ≤ 1 , but most comparator studies used in the MAICs enrolled patients with ECOG PS ≤ 2 . Moreover, GO29781 enrolled patients with Grade 1-3a FL, whereas the tazemetostat study enrolled patients with *EZH2*-wild-type Grade 3b or transformed FL. The comparator data in our MAICs versus duvelisib, umbralisib, and rituximab plus lenalidomide were based on patients with RR FL, of whom $\approx 12\%$ to 53% received 2L treatment, rather than those with RR FL who had received ≥ 2 prior therapies as in GO29781. Additionally, the data used for the MAIC versus rituximab plus lenalidomide were based on patients who were not rituximab refractory, some of whom had received only 1 prior regimen (53%), whereas the GO29781 FL cohort included 79% rituximab-refractory patients, and all had received ≥ 2 prior therapies, with $\geq 50\%$ having received ≥ 3 regimens. Given the

prognostic importance of these factors, these limitations may have introduced a bias in the results, which may have been particularly unfavorable for mosunetuzumab in those comparisons including patients in 2L or not refractory to rituximab, especially with regard to PFS and OS.

It was not always possible to adjust for all known prognostic factors and effect modifiers. In some cases, compromises had to be made to maximize the bias-variance tradeoff, which may inevitably have biased the results in an unknown direction and magnitude. Furthermore, because of suboptimal overlap between study populations, ESSs were relatively small in many ITCs. The small ESSs after adjustment in some comparisons, together with the low number of events observed for some endpoints, led to wide CIs and thus uncertain estimates. Notably, OS trends from these ITCs should be interpreted with caution, due to the limited number of events observed during the relatively short follow-up time in the mosunetuzumab and other comparator studies.

The matching and/or weighting adjustments conducted as part of the PSAs resulted at times in residual imbalances for multiple prognostic factors (Supplemental Results). Although these might have been further controlled for in subsequent outcome analyses, it is important to note that the second adjustment could only be performed for HRs and ORs but not for Kaplan-Meier curves, which should therefore be interpreted with care. Additionally, in some ITCs, OS and PFS survival curves crossed at 1 or more points, suggesting that the proportional hazard assumption may not hold and that the relative HRs should be interpreted in the context of the relevant Kaplan-Meier curves in the Supplemental results.

Endpoint definitions were also misaligned between GO29781 and some comparator studies. For example, ORR and CR results for the comparisons versus axicabtagene ciloleucel, tisagenlecleucel, umbralisib and rituximab plus bendamustine should be interpreted taking into account that tumor responses were assessed using the Lugano criteria²⁷ versus the International Harmonization Project 2007 revised response criteria³⁷ in GO29781. Moreover, as the incidence of discontinuations due to AEs associated with both agents was not available for the comparison versus rituximab plus lenalidomide, the comparison was based only on the proportion of patients who discontinued lenalidomide due to AEs. This is an underestimation of the tolerability of this treatment combination; therefore, the results from these comparisons are inevitably biased in favor of rituximab plus lenalidomide.

Despite these limitations, this study has several strengths. A large number of prognostic factors were considered in the ITCs to limit residual bias as much as possible. Multiple publications were reviewed not only to identify the comparator studies and patient cohorts that were most compatible with GO29781 to conduct an ITC but also to ensure the use of as many baseline characteristics as possible. Baseline characteristic and outcome measure definitions were aligned across trials whenever possible and appropriate, to further reduce bias in downstream estimates. Finally, sensitivity analyses were conducted to test the robustness of the base-case results and in general, and the conclusions were consistent across analyses.

Conclusions

To provide an overview of the comparative evidence versus standard-of-care and emerging therapies for RR FL after ≥ 2 prior therapies, a series of ITCs were conducted based on the available evidence to compare the effectiveness and tolerability of mosunetuzumab with key comparators of interest. The findings from this study suggest that mosunetuzumab could be a promising treatment option for patients with RR FL who have received ≥ 2 prior therapies. In addition to high rates of durable responses and a manageable safety profile, mosunetuzumab treatment also features other aspects such as a chemotherapy-free nature, fixed duration, overall ease of access and use in an outpatient setting, which are normally not investigated in the context of indirect treatment comparisons. Future studies with additional safety data from randomized trials or real-world studies would provide further understanding on the efficacy and safety profile of these therapies.

Clinical Practice Points

- Mosunetuzumab may lead to superior outcomes over PI3K inhibitors and tazemetostat in patients with RR FL without *EZH2* mutations and those without satisfactory alternative treatment options after ≥ 2 prior lines of treatment, without compromising on tolerability.
- A clear and consistent differentiation between mosunetuzumab and anti-CD20 antibody-based regimens as a class cannot be established across all efficacy and tolerability outcomes with the currently available evidence base. Nevertheless, bispecific antibodies may offer a promising alternative to re-challenging with a different anti-CD20 regimen in patients whose disease has relapsed or who did not respond to such treatments.
- Some efficacy outcomes with CART therapies might potentially be superior to those with mosunetuzumab, although important caveats and uncertainties still persist owing to intrinsic differences in study designs, and the evidence in support of tisagenlecleucel is considerably more uncertain.

Availability of Data and Materials

For eligible studies qualified researchers may request access to individual patient-level clinical data through a data request platform. At the time of writing this request platform is Vivli, at <https://vivli.org/ourmember/roche/>. For up-to-date details on Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents, see https://go.roche.com/data_sharing.

Authors' Contribution

Danilo Di Maio and Marie-Helene Blanchet Zumofen developed the study concept and design. Danilo Di Maio conducted the literature searches, analyses and interpretation of data. Danilo Di Maio, Michael C. Wei, Marie-Helene Blanchet Zumofen, Francesc Bosch, John Kuruvilla, Theodoros P. Vassilakopoulos and Loretta J. Nastoupil drafted the manuscript. Danilo Di Maio, Michael C. Wei, Marie-Helene Blanchet Zumofen, Francesc Bosch, John Kuruvilla, Theodoros P. Vassilakopoulos and Loretta J. Nastoupil conducted critical revision of the paper for important intellectual content.

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Francesc Bosch provided supervision. All authors read and approved the final manuscript.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.clml.2023.09.007.

References

1. Swerdlow SH, Campo E, Harris NL, et al. *WHO classification of tumours of haematopoietic and lymphoid tissues*. Switzerland: WHO Press, World Health Organization; 2017.
2. Swerdlow SH, Campo E, Pileri SA, et al. The 2016 revision of the World Health Organization classification of lymphoid neoplasms. *Blood*. 2016;127:2375–2390.
3. Dreyling M, Ghielmini M, Rule S, et al. Newly diagnosed and relapsed follicular lymphoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Ann Oncol*. 2021;32:298–308.
4. Carbone A, Roulland S, Gloghini A, et al. Follicular lymphoma. *Nat Rev Dis Primers*. 2019;5:83.
5. Casulo C, Larson MC, Lunde JJ, et al. Treatment patterns and outcomes of patients with relapsed or refractory follicular lymphoma receiving three or more lines of systemic therapy (LEO CREWE): a multicentre cohort study. *Lancet Haematol*. 2022;9:e289–e300.
6. Gopal AK, Kahl BS, de Vos S, et al. PI3K δ inhibition by idelalisib in patients with relapsed indolent lymphoma. *N Engl J Med*. 2014;370:1008–1018.
7. Flinn IW, Miller CB, Ardesna KM, et al. DYNAMO: duvelisib (IPI-145) in patients with refractory indolent non-Hodgkin lymphoma. *J Clin Oncol*. 2019;37:912–922.
8. Fowler NH, Samaniego F, Jurczak W, et al. Umbralisib, a dual PI3K δ /CK1 ϵ inhibitor in patients with relapsed or refractory indolent lymphoma. *J Clin Oncol*. 2021;39:1609–1618.
9. Chauhan AF, Cheson BD. Copanlisib in the treatment of relapsed follicular lymphoma: utility and experience from the clinic. *Cancer Manag Res*. 2021;13:677–692.
10. Richardson NC, Kasamon Y, Pazdur R, Gormley N. The saga of PI3K inhibitors in haematological malignancies: survival is the ultimate safety endpoint. *Lancet Oncol*. 2022;23:563–566.
11. Jacobson CA, Chavez JC, Sehgal AR, et al. Axicabtagene ciloleucel in relapsed or refractory indolent non-Hodgkin lymphoma (ZUMA-5): a single-arm, multicentre, phase 2 trial. *Lancet Oncol*. 2022;23:91–103.
12. Schuster SJ, Dickinson MJ, Dreyling MH, et al. Efficacy and safety of tisagenlecleucel in adult patients with relapsed/refractory follicular lymphoma: primary analysis of the phase 2 ELARA trial. *J Clin Oncol*. 2021;31:7508 Abstract.
13. Morschhauser F, Tilly H, Chaidos A, et al. Tazemetostat for patients with relapsed or refractory follicular lymphoma: an open-label, single-arm, multicentre, phase 2 trial. *Lancet Oncol*. 2020;21:1433–1442.
14. Budde LE, Assouline S, Sehn LH, et al. Single-agent mosunetuzumab shows durable complete responses in patients with relapsed or refractory B-cell lymphomas: phase I dose-escalation study. *J Clin Oncol*. 2022;40:481–491.
15. U.S. Food and Drug Administration. FDA grants accelerated approval to mosunetuzumab-axgb for relapsed or refractory follicular lymphoma 2022. <https://www.fda.gov/drugs/resources-information-approved-drugs/fda-grants-accelerated-approval-mosunetuzumab-axgb-relapsed-or-refractory-follicular-lymphoma>. Accessed February 2, 2023.
16. European Medicines Agency. Lunsumio [summary of product characteristics]. Roche Pharma AG. 2022.
17. Kang C. Mosunetuzumab: first approval. *Drugs*. 2022;82:1229–1234.
18. Sun LL, Ellerman D, Mathieu M, et al. Anti-CD20/CD3 T cell-dependent bispecific antibody for the treatment of B cell malignancies. *Sci Transl Med*. 2015;7:287ra70.
19. Signorovitch JE, Sikirica V, Erder MH, et al. Matching-adjusted indirect comparisons: a new tool for timely comparative effectiveness research. *Value Health*. 2012;15:940–947.
20. Phillippo DM, Ades AE, Dias S, Palmer S, Abrams KR, Welton NJ. Methods for population-adjusted indirect comparisons in health technology appraisal. *Med Decis Making*. 2018;38:200–211.
21. Faria R, Hernandez Alava M, Manca A, Walloo AJ. *NICE DSU Technical Support Document 17: the use of observational data to inform estimates of treatment effectiveness for technology appraisal: methods for comparative individual patient data*; 2015. <https://www.sheffield.ac.uk/sites/default/files/2022-02/TSD17-DSU-Observational-data-FINAL.pdf>. Accessed February 18, 2022.
22. Austin PC, Small DS. The use of bootstrapping when using propensity-score matching without replacement: a simulation study. *Stat Med*. 2014;33:4306–4319.
23. Jacobson C, Chavez JC, Sehgal AR, et al. Primary analysis of ZUMA-5: a phase 2 study of axicabtagene ciloleucel (AXI-CEL) in patients with relapsed/refractory (R/R) indolent non-hodgkin lymphoma (INHL). *Blood*. 2020;136(suppl 1):40–41.
24. Jacobson C, Chavez JC, Sehgal AR, et al. Interim analysis of ZUMA-5: a phase II study of axicabtagene ciloleucel (axi-cel) in patients (pts) with relapsed/refractory indolent non-Hodgkin lymphoma (R/R iNHL). *J Clin Oncol*. 2020;38(suppl 15):8008 Abstract.
25. Neelapu SS, Chavez JC, Sehgal AR, et al. Long-term follow-up analysis of ZUMA-5: a phase 2 study of axicabtagene ciloleucel (axi-cel) in patients with relapsed/refractory indolent non-Hodgkin lymphoma. *Blood*. 2021;138(suppl 1):93.
26. Palomba ML, Ghione P, Patel AR, et al. A comparison of clinical outcomes from updated ZUMA-5 (axicabtagene ciloleucel) and the international SCHOLAR-5 external control cohort in relapsed/refractory follicular lymphoma (R/R FL). *Blood*. 2021;138(suppl 1):3543.
27. Cheson BD, Fisher RI, Barrington SF, et al. Recommendations for initial evaluation, staging, and response assessment of Hodgkin and non-Hodgkin lymphoma: the Lugano classification. *J Clin Oncol*. 2014;32:3059–3068.
28. Hao Y, Hsu W-C, Parzynski CS, et al. Comparison of clinical outcomes among patients with relapsed/refractory follicular lymphoma treated with tisagenlecleucel in the ELARA trial versus a real-world external control arm of patients treated with standard of care. *Blood*. 2021;138(suppl 1):2419.
29. Salles G, Schuster SJ, Dreyling MH, et al. Efficacy comparison of tisagenlecleucel versus standard of care in patients with relapsed or refractory follicular lymphoma. *Blood*. 2021;138(suppl 1):3528.

30. Thieblemont C, Dickinson M, Martinez-Lopez J, et al. Efficacy of tisagenlecleucel in adult patients with high-risk relapsed/refractory follicular lymphoma: subgroup analysis of the phase II ELARA study. *Blood*. 2021;138(suppl 1):131.
31. Fowler NH, Dickinson M, Dreyling M, et al. Tisagenlecleucel in adult relapsed or refractory follicular lymphoma: the phase 2 ELARA trial. *Nat Med*. 2022;28:325–332.
32. Fowler NH, Dickinson M, Ghosh M, et al. Assessment of healthcare resource utilization and costs in patients with relapsed or refractory follicular lymphoma undergoing CAR-T cell therapy with tisagenlecleucel: results from the ELARA study. *Blood*. 2021;138(suppl 1):3533.
33. Eyre TA, Osborne WL, Gallop-Evans E, et al. Results of a multicentre UK-wide compassionate use programme evaluating the efficacy of idelalisib monotherapy in relapsed, refractory follicular lymphoma. *Br J Haematol*. 2018;181:555–559.
34. Gopal AK, Kahl BS, Flowers CR, et al. Idelalisib is effective in patients with high-risk follicular lymphoma and early relapse after initial chemioimmunotherapy. *Blood*. 2017;129:3037–3039.
35. Salles G, Schuster SJ, de Vos S, et al. Efficacy and safety of idelalisib in patients with relapsed, rituximab- and alkylating agent-refractory follicular lymphoma: a subgroup analysis of a phase 2 study. *Haematologica*. 2017;102:e156–e159.
36. Wagner-Johnston ND, Schuster SJ, deVos S, et al. Outcomes of patients with up to 6 years of follow-up from a phase 2 study of idelalisib for relapsed indolent lymphomas. *Leuk Lymphoma*. 2021;62:1077–1087.
37. Cheson BD, Pfistner B, Juweid ME, et al. Revised response criteria for malignant lymphoma. *J Clin Oncol*. 2007;25:579–586.
38. European Medicines Agency. *Copiktra Assessment Report (EPAR)*. European Medicines Agency; 2021.
39. Zinzani PL, Samaniego F, Jurczak W, et al. Umbralisib, the once daily dual inhibitor of PI3K δ and casein kinase-1 ϵ demonstrates clinical activity in patients with relapsed or refractory indolent non-Hodgkin lymphoma: results from the phase 2 global Unity-NHL trial. *Blood*. 2020;136(suppl 1):34–35.
40. Leonard JP, Trneny M, Izutsu K, et al. AUGMENT: a phase III study of lenalidomide plus rituximab versus placebo plus rituximab in relapsed or refractory indolent lymphoma. *J Clin Oncol*. 2019;37:1188–1199.
41. European Medicines Agency. *Revlimid Assessment Report (EPAR)*. European Medicines Agency; 2022.
42. Leonard JP, Trneny M, Izutsu K, et al. AUGMENT phase III study: lenalidomide/rituximab (R²) improved efficacy over rituximab/placebo in relapsed/refractory follicular patients irrespective of POD24 status. *Hematol Oncol*. 2019;37:114–115.
43. Salles G, Tilly H, Chaidos A, et al. Analyzing efficacy outcomes from the phase 2 study of single-agent tazemetostat as third-line therapy in patients with relapsed or refractory follicular lymphoma to identify predictors of response. *Blood*. 2020;136(suppl 1):47–49.
44. Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. *Stat Med*. 2015;34:3661–3679.
45. Rosenbaum PR, Rubin DB. The central role of the propensity score in observational studies for causal effects. *Biometrika*. 1983;70:41–55.
46. Hansen BB, Klopfer SO. Optimal full matching and related designs via network flows. *J Comput Graph Stat*. 2006;15:606–627.
47. Diamond A, Sekhon JS. Genetic matching for estimating causal effects: a general multivariate matching method for achieving balance in observational studies. *Rev Econ Stat*. 2013;95:932–945.
48. Austin PC, Stuart EA. Optimal full matching for survival outcomes: a method that merits more widespread use. *Stat Med*. 2015;34:3949–3967.
49. Franklin JM, Rassen JA, Ackermann D, Bartels DB, Schneeweiss S. Metrics for covariate balance in cohort studies of causal effects. *Stat Med*. 2014;33:1685–1699.
50. Stuart EA, Lee BK, Leacy FP. Prognostic score-based balance measures can be a useful diagnostic for propensity score methods in comparative effectiveness research. *J Clin Epidemiol*. 2013;66:S84–S90.e81.
51. Austin PC. The performance of different propensity score methods for estimating marginal odds ratios. *Stat Med*. 2007;26:3078–3094.
52. Austin PC, Stuart EA. Estimating the effect of treatment on binary outcomes using full matching on the propensity score. *Stat Methods Med Res*. 2017;26:2505–2525.
53. Schuster S, Dickinson M, Dreyling M, et al. Efficacy and safety of tisagenlecleucel (Tisa-cel) in adult patients (Pts) with relapsed/refractory follicular lymphoma (r/r FL): primary analysis of the phase 2 Elara trial. *J Clin Oncol*. 2021;39:7508.
54. Sehn LH, Chua N, Mayer J, et al. Obinutuzumab plus bendamustine versus bendamustine monotherapy in patients with rituximab-refractory indolent non-hodgkin lymphoma (GADOLIN): a randomised, controlled, open-label, multicentre, phase 3 trial. *Lancet Oncol*. 2016;17:1081–1093.
55. Sehn LH, Goy A, Offner FC, et al. Randomized phase II trial comparing obinutuzumab (GA101) with rituximab in patients with relapsed CD20+ indolent B-cell non-hodgkin lymphoma: final analysis of the GAUSS study. *J Clin Oncol*. 2015;33:3467–3474.
56. Sehn LH, Kamdar M, Herrera AF, et al. Randomized phase 2 trial of polatuzumab vedotin (pola) with bendamustine and rituximab (BR) in relapsed/refractory (r/r) FL and DLBCL. *J Clin Oncol*. 2018;36(suppl 15):7507 Abstract.
57. van Oers MH, Klasa R, Marcus RE, et al. Rituximab maintenance improves clinical outcome of relapsed/resistant follicular non-Hodgkin lymphoma in patients both with and without rituximab during induction: results of a prospective randomized phase 3 intergroup trial. *Blood*. 2006;108:3295–3301.
58. Zinzani PL, Flinn IW, Yuen SLS, et al. Venetoclax-rituximab with or without bendamustine vs bendamustine-rituximab in relapsed/refractory follicular lymphoma. *Blood*. 2020;136:2628–2637.
59. Proudman D, Nellesen D, Gupta D, Adib D, Yang J, Mamlouk K. A matching-adjusted indirect comparison of single-arm trials in patients with relapsed or refractory follicular lymphoma who received at least two prior systemic treatments: tazemetostat was associated with a lower risk for safety outcomes versus the PI3-kinase inhibitors idelalisib, duvelisib, copanlisib, and umbralisib. *Adv Ther*. 2022;39:1678–1696.
60. Dickinson M, Martinez-Lopez J, Jousseau E, et al. Comparative efficacy and safety of tisagenlecleucel and axicabtagene ciloleucel in relapsed/refractory follicular lymphoma. *Transplant Cell Ther*. 2022;28:S65–S66.